

Task Questions

1. Question 1

After I downloaded the files, I unzipped the files and analyzed the files inside it. I found out that those folders have the python codes and packages that are used by the Lambda functions

2. Question 2

So that the Lambda function can be able to communicate with the RDS instance

3. Question 3

Yes, it can be stored as an AWS system manager parameter instead of an environment variable

4. Question 4

No, we will not receive any further emails about the notifications whenever there is a trigger to be notified if we don't confirm the topic subscription.

5. Question 5

We can use the logs from the amazon cloud watch logs, and we will be able to see where exactly the error started from. So, after finding out about that, we can resolve the issue from the information we get from the logs.

Module 13 Challenge Lab - Implementing a Serverless Architecture for the Cafe

Task 1: Downloading the source code

first, I downloaded the two zip files containing the code for generating the report. I examined the contents to understand what they're all about.

Name	Date modified	Type	Size
salesAnalysisReport	6/2/2024 11:45 PM	File folder	
salesAnalysisReportDataExtractor	6/2/2024 11:46 PM	File folder	

These files likely include Python scripts and possibly other resources needed by the program. While I was at it, I answered a question about the code structure.

AWS Academy Cloud Architecting - Module 13 Challenge Lab Questions

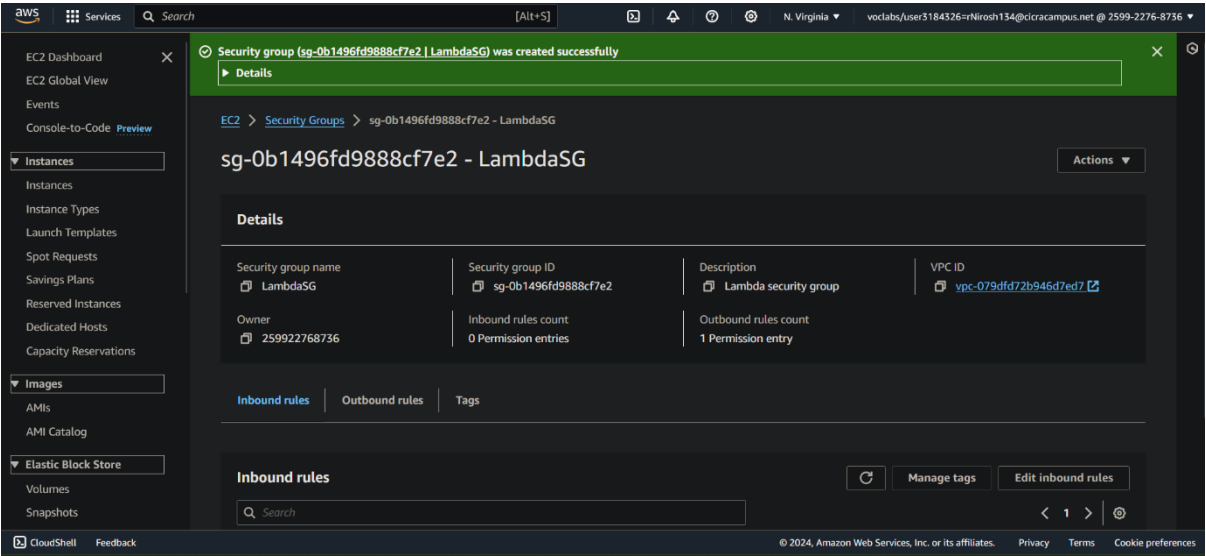
View questions in: [English](#)

Question 1: Why does the salesAnalysisReportDataExtractor.zip file have a package folder?

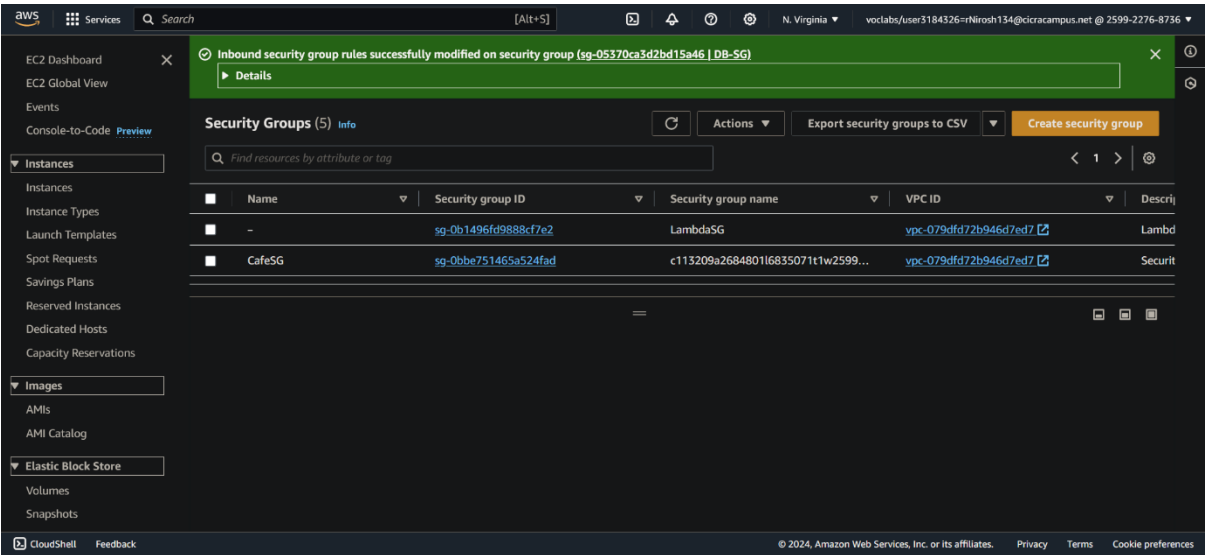
- ☐ It's an optional folder to improve caching within the Lambda function
- ☒ The folder contains any Python packages that are used by the Lambda function
- ☐ The folder contains debugging information for Python
- ☐ The folder is a required folder for Lambda functions that are deployed to a VPC

Task 2: Creating the *DataExtractor* Lambda function in the VPC

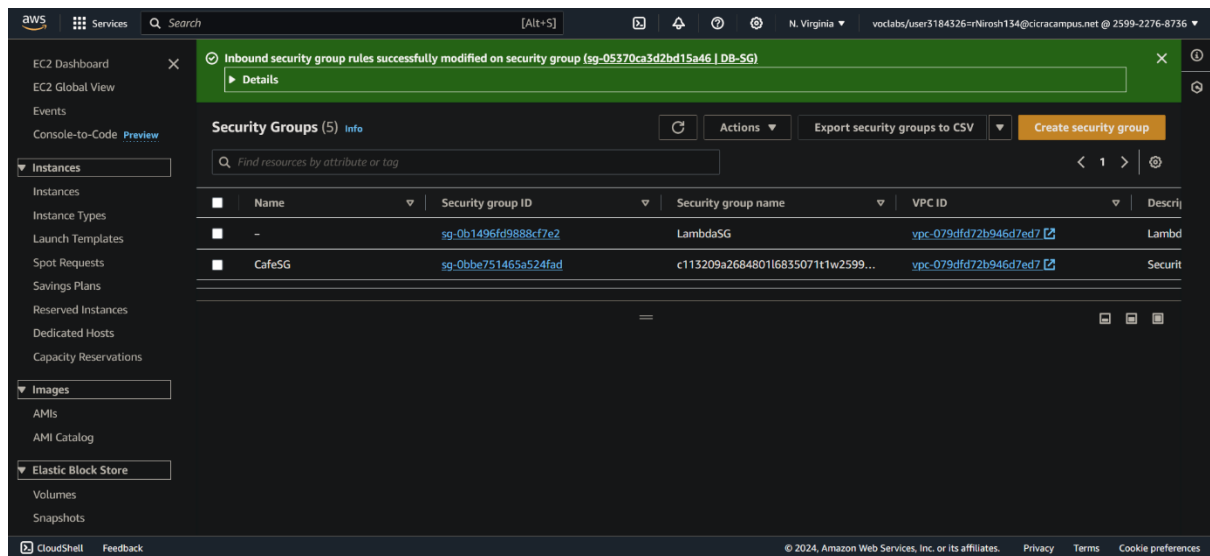
I built a special function called "salesAnalysisReportDataExtractor" to grab the sales info from the cafe's database.



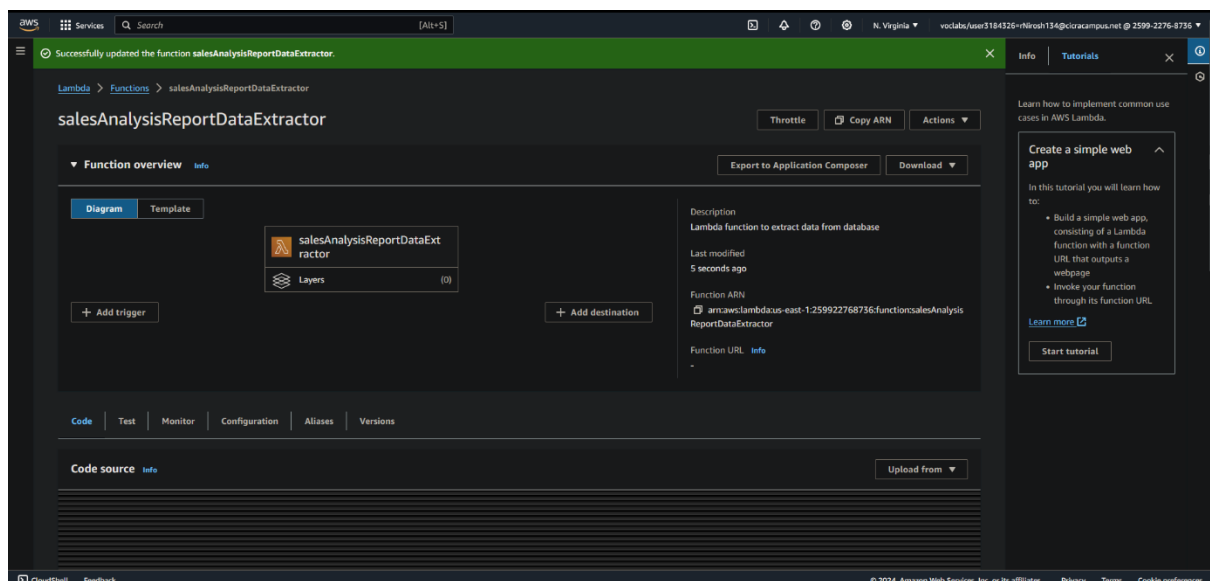
To ensure secure access, I created a security group that acts like a digital gatekeeper for the function to communicate with the database.



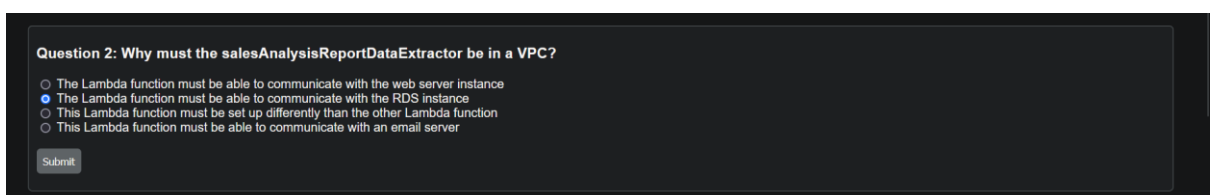
Then, I configured the function itself, specifying things like the programming language (Python 3.8 in this case) and memory allocation.



Finally, I uploaded the code from the downloaded zip file containing the data extraction logic.

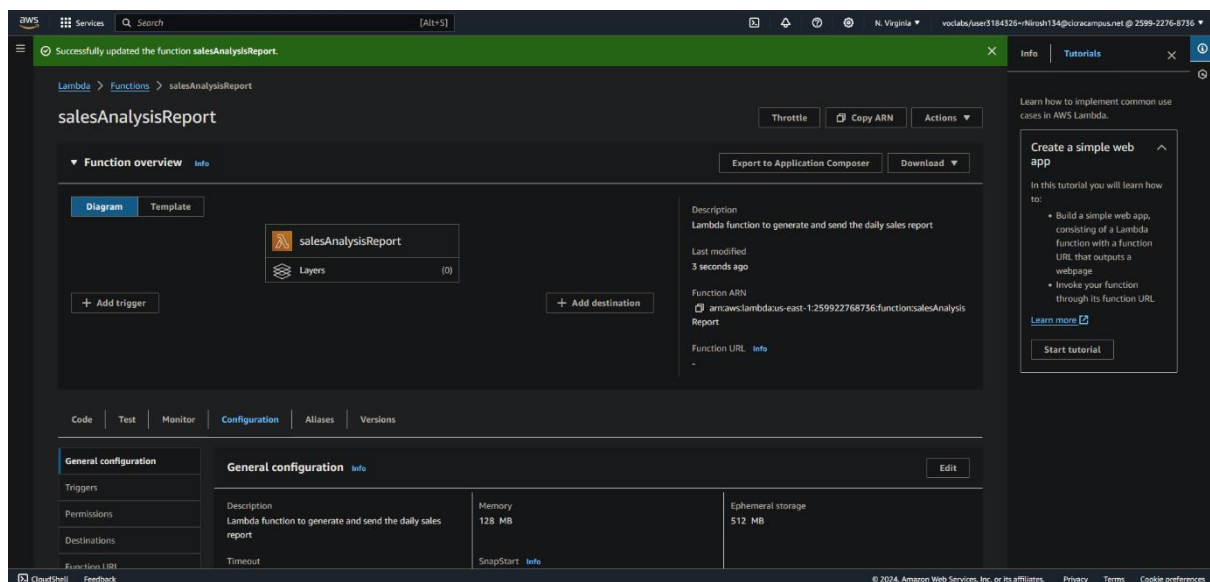
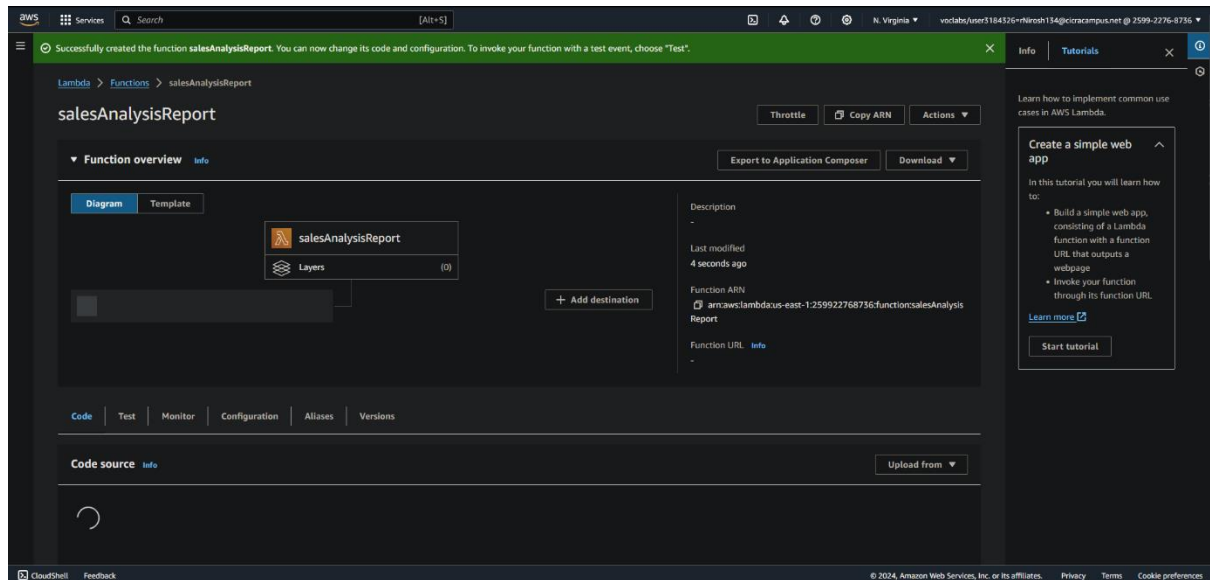


There was a question here about why this function needs to be within a specific virtual network (VPC)



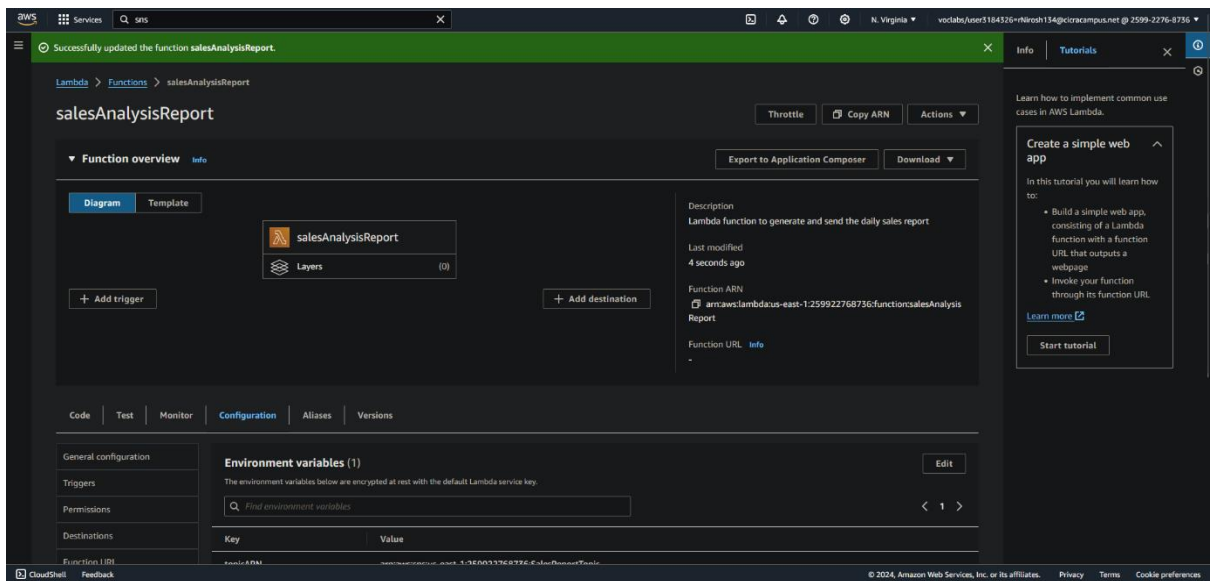
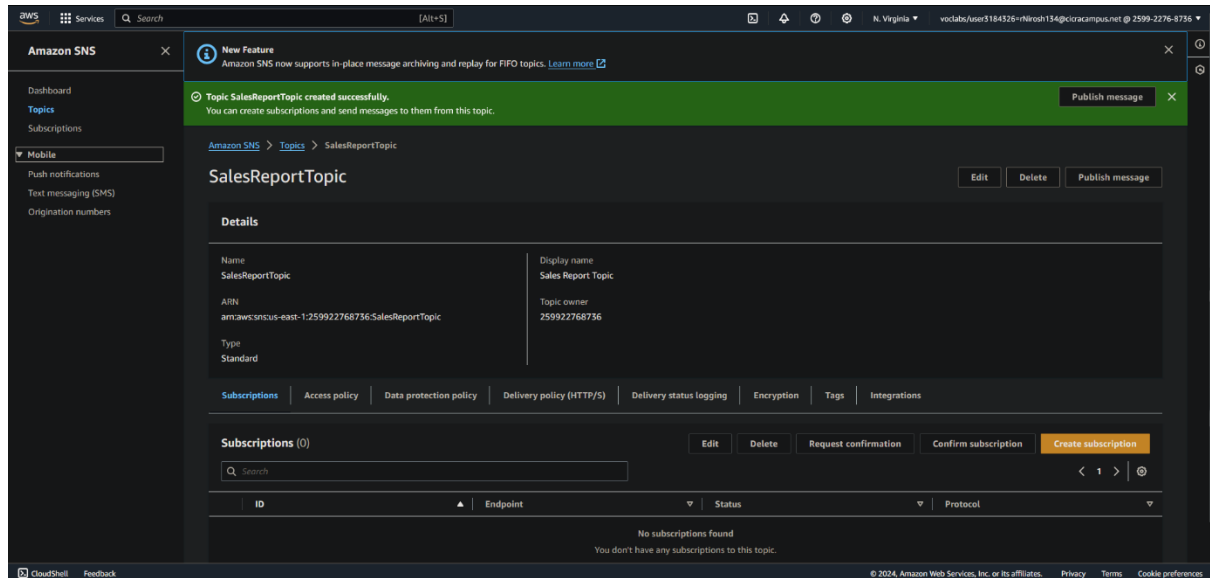
Task 3: Creating the *salesAnalysisReport* Lambda function

Next, I created another Lambda function named "salesAnalysisReport" to take care of generating the report itself. It used the Python code from the other downloaded zip file and had similar settings to the data extractor function.



Task 4: Creating an SNS topic

To send out the report, I created a special notification service called an "SNS topic" named "SalesReportTopic". This topic acts like a central hub for sending messages. I grabbed its unique identifier (ARN) to configure the report generation function later.



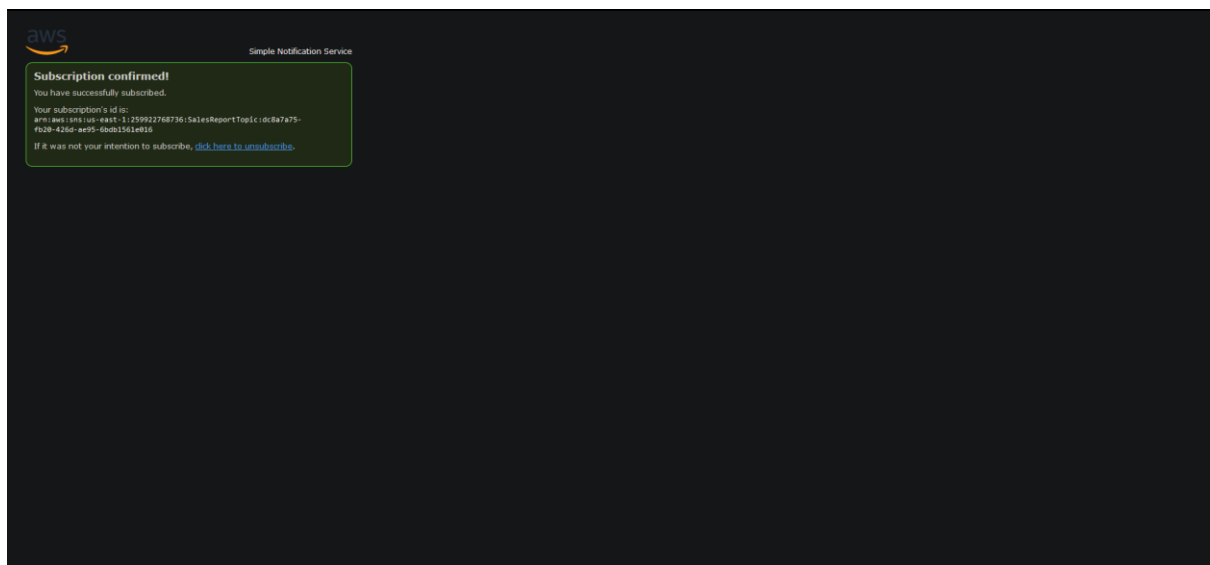
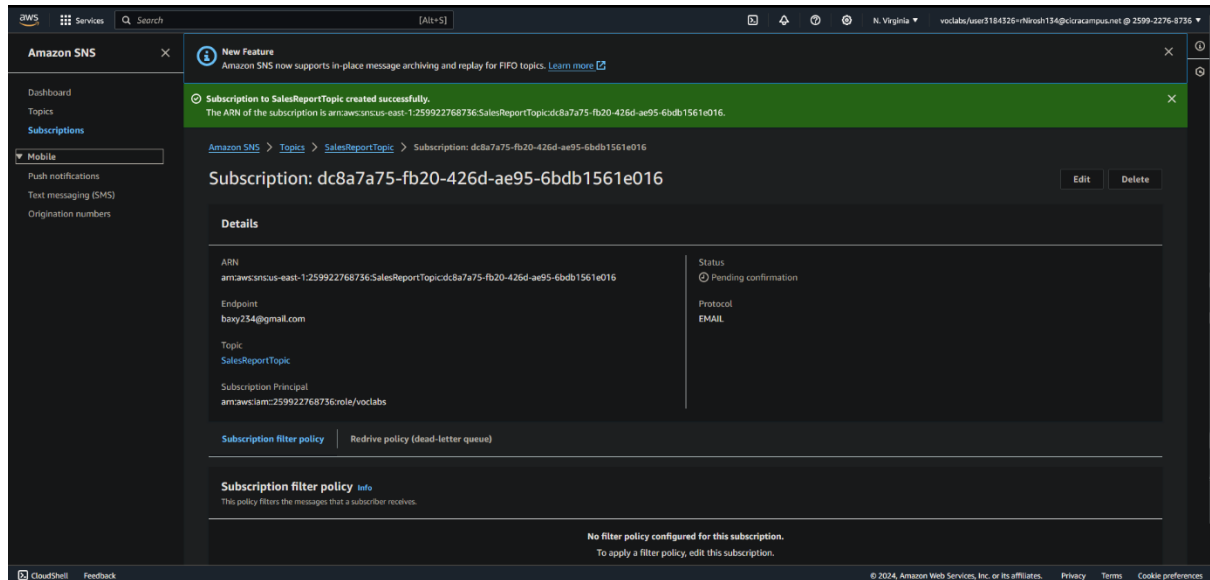
Question 3: Could the topicARN be stored as an AWS Systems Manager parameter instead of as an environment variable (assuming that the code could be updated)?

- ☒ Yes
☐ No

Submit

Task 5: Creating an email subscription to the SNS topic

To get the report delivered, I subscribed my email address to the SNS topic I just created. This way, whenever the report is generated, it would be sent to my inbox.



Question 4: Will you receive an email message if you do not confirm the topic subscription?

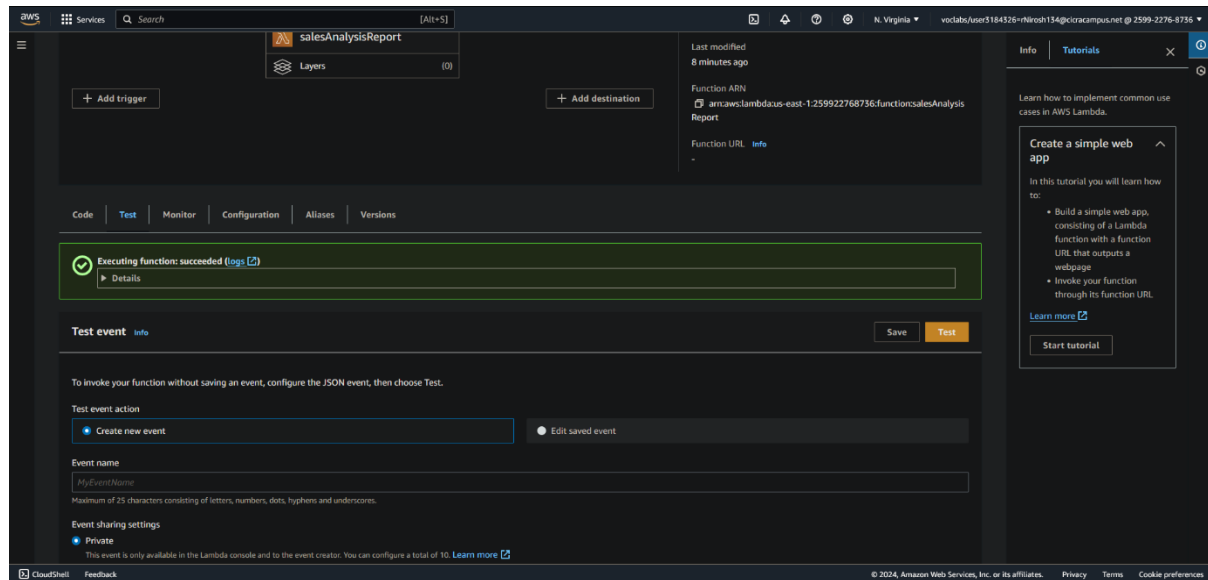
☐ Yes

☒ No

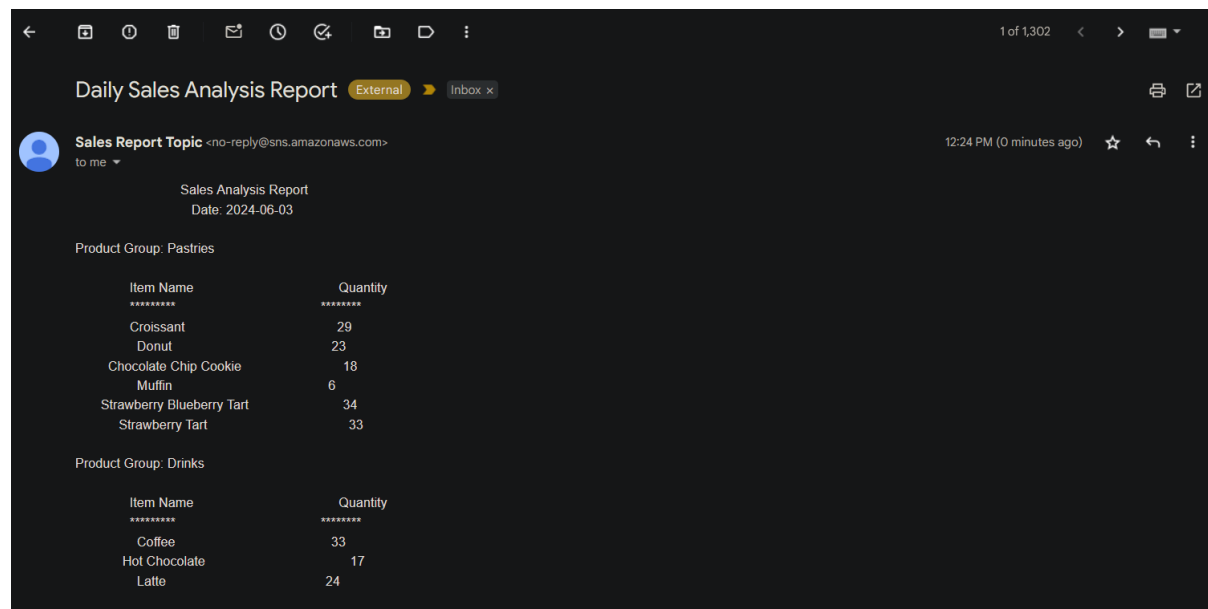
Submit

Task 6: Testing the *salesAnalysisReport* Lambda function

I made sure everything worked. I created a test for the report generation function and ran it. If all went well, I should have received a test report via email in a few minutes. If not, there were some troubleshooting tips that could help me figure out what went wrong.

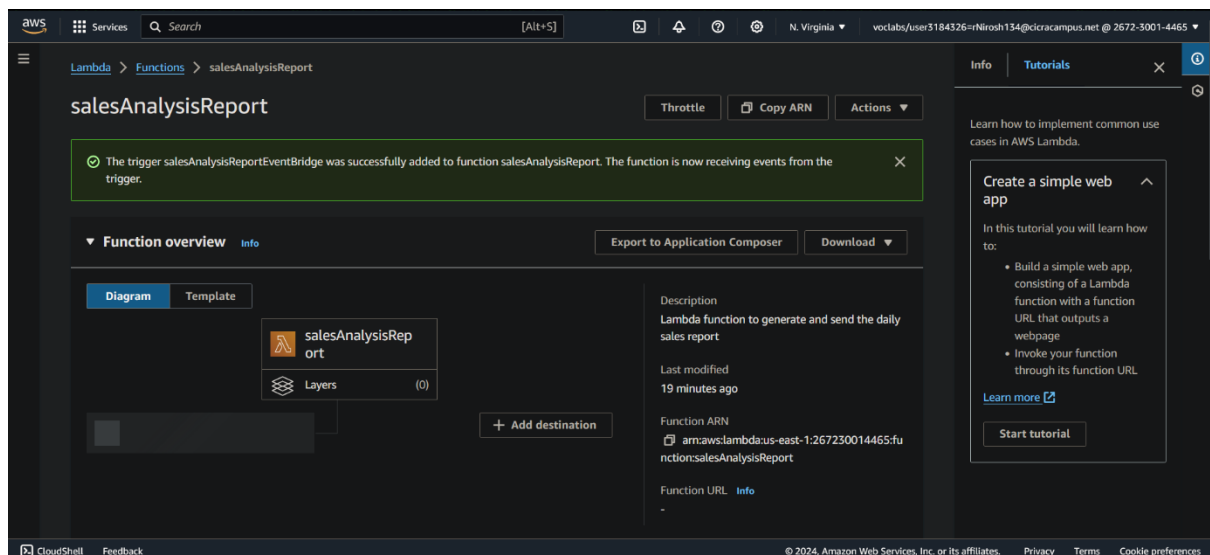


Then I got the email notifying me about the test.

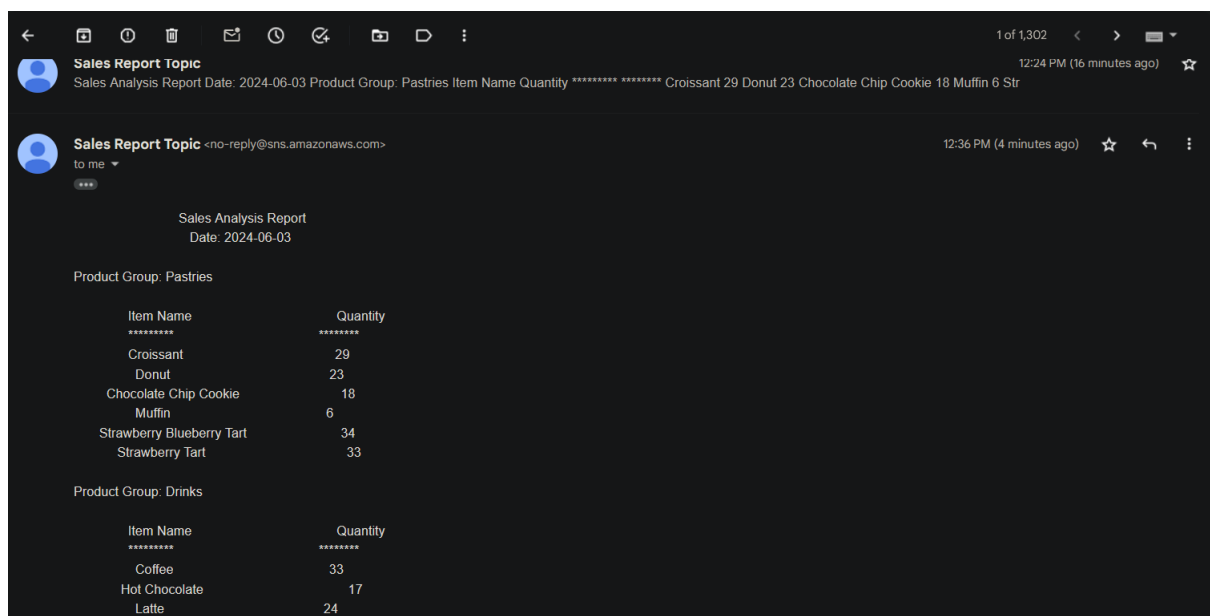


Task 7: Setting up an Amazon EventBridge event to trigger the Lambda function each day

Finally, to automate the whole process, I set up a scheduler called an "EventBridge rule" to trigger the report generation function every day at a specific time. They wanted me to use a special code (cron expression) to define the timing – I made sure it was in Coordinated Universal Time (UTC).



After that, I checked my email again to see if the scheduled report arrived. It arrived.



Question 5: Frank tells you that he hasn't received an email report in the last few days. What could you do to troubleshoot this issue?

- ☐ Restart the Lambda function because it might be stuck
- ☐ Update the Python version
- ☒ Use the logs from Amazon CloudWatch Logs and review them for errors
- ☐ Use the AWS CloudTrail logs and review them for errors

Submit

SIT233 – Cloud Computing
Module 13 Challenge Lab - Implementing a Serverless
Architecture for the Cafe

Name: Nirosh Ravindran
CICRA Student ID: BSCP|CS|63|134

Grade:

My Grade is 35/35

The screenshot shows the AWS Academy interface for a challenge lab. The left sidebar contains navigation links: Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area is titled 'Module 13 Challenge Lab - Implementing a Serverless Architecture for the Cafe'. Below the title, it shows 'Due No Due Date', 'Points 100', and 'Submitting an external tool'. A toolbar includes buttons for 'Submit', 'Details', 'AWS', 'Start Lab', 'End Lab', '2:02', 'Instructions', 'Grades', and 'Actions'. The 'Grades' button is active, displaying a 'Total score 35/35'. Below the score, a list of tasks is shown: '[Task 2] Lambda security group exists', '[Task 2] Lambda function salesAnalysis', '[Task 3] Lambda function salesAnalysis', '[Task 4] SNS topic exists', and '[Task 5] Email subscription exists'. The 'Submitting your work' section contains instructions for submitting progress and a tip about multiple submissions. A terminal window is open, showing a bash prompt and the output of a command: 'Default region: us-east-1', 'Back in submit.sh...', and 'end'.

aws academy

ACAv2EN... > Assignments
> Module 13 Challenge Lab - Implementing a Serverless Architecture for the Cafe

Module 13 Challenge Lab - Implementing a Serverless Architecture for the Cafe

Due No Due Date Points 100 Submitting an external tool

Submit Details AWS Start Lab End Lab 2:02 Instructions Grades Actions

Files README Terminal Source

EN_US

Submitting your work

28. At the top of these instructions, choose **Submit** to record your progress and when prompted, choose **Yes**.

29. If the results don't display after a couple of minutes, return to the top of these instructions and choose **Grades**.

Tip: You can submit your work multiple times. After you change your work, choose **Submit** again. Your last submission is what will be recorded for this lab.

30. To find detailed feedback on your work, choose **Details** followed by **View Submission Report**.

```
bash
$ cd /mnt/vocwork3/grades/eee_6_3362885/asn2684880_18/asn2684880
$ /usr/lib/python3.7/site-packages/boto3/compat.py:82: Python
r support Python 3.7 starting December 13, 2023. To continue r
security updates please upgrade to Python 3.8 or later. More
ws.amazon.com/blogs/developer/python-support-policy-updates-fo
warnings:warn(warning: PythonDeprecationWarning)
Present working directory = /mnt/vocwork3/grades/eee_6_236
1/work
Default region: us-east-1
Back in submit.sh...
end
eee_W_3866282@runweb125872:~$
eee_W_3866282@runweb125872:~$
```

Total score 35/35

[Task 2] Lambda security group exists

[Task 2] Lambda function salesAnalysis

[Task 3] Lambda function salesAnalysis

[Task 4] SNS topic exists

[Task 5] Email subscription exists